

Section II: Remarks

Claim 12 is pending. Claim 12 has been amended herein, without prejudice. Support for the amendment to claim 12 can be found in paragraphs [0012] and [0017] of the present application.

No new matter is added.

Request for Rejoinder

Applicants respectfully remind the Examiner that upon allowance of a product claim, the Examiner can rejoin the withdrawn process claims 21-23 which are written in dependent form or otherwise include all of the limitations of the allowed product claim and fully examine the process claims for patentability, as provided by MPEP §821.04.

Rejection of the Claims

In the June 15, 2015 Office Action, the Examiner stated that:

claim 12 was rejected under 35 U.S.C. §112, first paragraph;

claim 12 was rejected under 35 U.S.C. §112, second paragraph;

claim 12 was rejected under 35 U.S.C. §101; and

claim 12 was rejected under 35 U.S.C. §103(a) as being unpatentable over “Rubb-e-Jamun Barae Zaibetus” (U1) in view of “Pata de Vaca” (V1), as evidenced by W1 (<https://web.archive.org/web/20010208172023/http://rain-tree.com/prepmethod.htm>).

Applicants traverse such rejections and request reconsideration in view of the following remarks.

Rejection under 35 U.S.C. §112

1. In the June 15, 2015 Office Action, claim 12 was rejected under 35 U.S.C. §112, first paragraph, as failing to comply with the written description requirement. Applicants traverse such rejection.

According to the Examiner, the limitations “about 382.5 mg” and “about 67.5 mg” are not supported in the as-filed specification. Although not acquiescing to the propriety of the Examiner’s rejection, applicants have amended claim 12 to recite:

12. A product to be administered to individuals suffering from type II diabetes, wherein said product comprises a mixture of ground and sterilized *Syzygium cumini* seeds and *Bauhinia forficata* leaves, wherein the percentage by weight of *Syzygium cumini* seeds is about 85% and the percentage by weight of *Bauhinia forficata* leaves is about 15%.

Support for the amendment to claim 12 can be found in the instant specification at paragraph [0012], which recites:

[0012] Within these percentages by weight satisfactory results have been reached with a percentage by weight of 85% of *Syzygium* and a percentage by weight of 15% of *Bauhinia*.

and paragraph [0017], which recites:

[0017] After a quality control process, the grinding phase is started, in which the seeds and the leaves are ground, after which they are sterilized.

With the amendment to claim 12, the rejection under 35 U.S.C. §112, first paragraph, as lacking written description is obviated. Withdrawal of the rejection is respectfully requested.

2. In the June 15, 2015 Office Action, claim 12 was rejected under 35 U.S.C. §112, second paragraph, as being indefinite for failing to particularly point out and distinctly claim what the applicants regard as the invention. Applicants traverse such rejection.

According to the Examiner, there is a lack of clarity with regards to the terms “An herbal product” and “blend.” As introduced hereinabove, claim 12 has been amended and with the amendment it is clear that applicants are claiming a composition. Withdrawal of the rejection is respectfully requested.

Rejection under 35 U.S.C. §101

In the June 15, 2015 Office Action, claim 12 was rejected under 35 U.S.C. §101 as allegedly being directed to patent ineligible subject matter. Applicants traverse such rejection.

According to the Examiner, applicants' claims 12-14 are drawn to be administered to individuals suffering from type II diabetes, said product comprising a blend of about 382.5 mg of *Syzygium cumini* seeds and about 67.5 mg of *Bauhinia forficata* leaves. As introduced hereinabove, claim 12 now recites a product said product comprising a mixture of ground and sterilized *Syzygium cumini* seeds and *Bauhinia forficata* leaves, wherein the percentage by weight of *Syzygium cumini* seeds is about 85% and the percentage by weight of *Bauhinia forficata* leaves is about 15%.

Since the claims involve a law of nature and involve a nature-based product, applicants must assess whether the nature-based product has markedly different characteristics relative to its naturally occurring counterpart. In this case, pursuant to the Guidelines, "when the nature-based product is produced by combining multiple components, the markedly different characteristics analysis should be applied to the resultant nature-based combinations, rather than its component parts." In other words, in this case, the product comprising comprising a mixture of ground and sterilized *Syzygium cumini* seeds and *Bauhinia forficata* leaves, wherein the percentage by weight of *Syzygium cumini* seeds is about 85% and the percentage by weight of *Bauhinia forficata* leaves is about 15% must be analyzed to determine if it has markedly different characteristics relative to its naturally occurring counterpart. In the present case, applicants have created a mixture that is not found in nature as said mixture. There is no objective reason to believe that the people that discovered that *Syzygium cumini* seeds can be used to treat diabetes and the people that discovered that *Bauhinia forficata* leaves could be used to balance blood levels in diabetics would combine a mixture of ground and sterilized *Syzygium cumini* seeds and *Bauhinia forficata* leaves, wherein the percentage by weight of *Syzygium cumini* seeds is about 85% and the percentage by weight of *Bauhinia forficata* leaves is about 15%, as claimed by applicants herein. There would be no way to predict that this ratio of *Syzygium cumini* seeds and *Bauhinia forficata* leaves can result in a synergistic lowered of blood glucose levels (see, for example, the Declaration under 37 CFR 1.132 in **Appendix A**).

Applicants again direct the Examiner to Example 10, claim 2, of the Nature-Based Products examples provided in December 2014, which was considered patent eligible.

2. A yogurt starter culture comprising: goat milk mixed with *Streptococcus thermophilus* and *Lactobacillus alexandrinus*.

Claim 2: Eligible. As described in the specification, when *S. thermophilus* and *L. alexandrinus* are mixed, the two bacterial species have different characteristics than either species does on its own, e.g., they act together to ferment milk into a lower fat yogurt than either bacteria can produce when individually mixed with the milk. Thus, the mixture of the bacteria and milk has different functional characteristics (lower fat content) than the naturally occurring bacteria (or milk) by itself. These differences rise to the level of a marked difference, and accordingly the claimed starter culture is not a "product of nature" exception. Thus, the claim is not directed to an exception (Step 2A: NO), and qualifies as eligible subject matter.

In the Example 10, claim 2 analysis, the two bacterial species *S. thermophilus* and *L. alexandrinus* display different characteristics than either species does on its own, e.g., they act together to ferment

mild into a lower fat yogurt than either bacteria can produce when individually mixed with the milk. This is analogous to applicants' presently claimed invention, whereby there is a synergistic lowering of blood glucose levels associated with the claimed ratio of *Syzygium cumini* seeds and *Bauhinia forficata* leaves. In other words, the *Syzygium cumini* seeds and *Bauhinia forficata* leaves act synergistically to lower the blood glucose levels to a value lower than either the *Syzygium cumini* seeds or the *Bauhinia forficata* leaves can do alone.

Accordingly, applicants' claimed product has markedly different characteristics than anything found in nature and as such, the claimed product is patent eligible. Withdrawal of the rejection of claim 12 under 35 U.S.C. §101 is respectfully requested.

Rejection under 35 U.S.C. §103(a)

In the June 15, 2015 Office Action, claim 12 was rejected under 35 U.S.C. §103(a) as being unpatentable over "Rubb-e-Jamun Barae Zaibetus" (U1) in view of "Pata de Vaca" (V1), as evidenced by W1 (<https://web.archive.org/web/20010208172023/http://rain-tree.com/prepmethod.htm>). Applicants traverse such rejection.

Applicants have amended claim 12 as introduced hereinabove.

With regards to the Examiner's use of *In re Kerkhoven*, in general, the effects of combination of two different drugs are inherently **unpredictable**, as to whether the net therapeutic effect will be additive, or synergistic, or even antagonistic. Further, it is well established that "[a] greater than expected result is an evidentiary factor pertinent to the legal conclusion of obviousness ... of the claims at issue." *In re Corkill*, 226 USPQ 1005 (Fed. Cir. 1985). Evidence of a greater than expected result may also be shown by demonstrating an effect which is greater than the sum of each of the effects taken separately (i.e., demonstrating "synergism"). *Merck & Co. Inc. v. Biocraft Laboratories Inc.*, 10 USPQ2d 1843 (Fed. Cir.), *cert. denied*, 493 U.S. 975 (1989).

Applicants submit herewith a new Declaration under 37 CFR §1.132 in the name of Juan Carlos AGREDA NAVAJAS (hereinafter the second Declaration). In the second Declaration, experiments were carried out to compare the blood glucose levels of metformin, ground and sterilized seeds of the *Syzygium cumini* plant (hereinafter referred to as JMB), ground and sterilized leaves of the *Bauhinia forficata* plant (hereinafter referred to as PV), and mixture of ground and sterilized *Syzygium cumini* seeds and *Bauhinia forficata* leaves at a ratio of 85:15 (hereinafter referred to as the "DMIND product") when administered to SHR hypertensive rats. The results clearly show a potential use for the plant extracts JMB, DMNID and PV in blood glucose levels reduction. All of them showed

significant blood glucose level lowering. Metformin appeared to have a similar effect, but the plants used had efficacy at lower dose (300 mg/kg and 200 mg/kg for metformin and the plants, respectively). Moreover, the DMNID, which is a mixture of JMB and PV at a ratio of 85:15, exhibited a clear synergic lowering of the glucose levels. Accordingly, the combination of *Syzygium cumini* seeds and *Bauhinia forficata* leaves at the claimed ratio synergistically lowered blood glucose levels in hypertensive rats.

If a *prima facie* case of obviousness is established, which applicants are not in any way acquiescing to, the burden shifts to the applicant to come forward with arguments and/or evidence to rebut the *prima facie* case. See, e.g., *In re Dillon*, 16 USPQ2d 1897, 1901 (Fed. Cir. 1990). Rebuttal evidence may consist of a showing that the claimed compound possesses unexpected properties. *Dillon*, 16 USPQ2d at 1901. In the present case, applicants have provided evidence in the second Declaration under 37 CFR §1.132 that the combination of *Syzygium cumini* seeds and *Bauhinia forficata* leaves in the specifically claimed ratio results in a product that statistically and synergistically lowers the blood glucose levels in hypertensive rats. Any allegations that it would have been *prima facie* obvious and expected to provide an even more effective formulation through combining prior art teachings directed to the same purpose, is respectfully traversed since it was impossible to predict the aforementioned unexpected synergistic effect provided by the presently claimed invention.

It is noted that the Examiner indicated in the June 15, 2015 Office Action that it would be an obvious starting point to one of ordinary skill in the art to combine equal amounts (i.e., about 50% by weight of each plant) of the plants and adjust the amounts of each plant in order to achieve the desired therapeutic effect. Applicants respectfully disagree with the alleged simplicity of this approach. There are hundreds of possible combinations of *Syzygium cumini* seeds and *Bauhinia forficata* leaves that can be envisioned by the person skilled in the art, amounting to hundreds of hours of work. To conclude that it would be obvious to determine that the claimed ratio is easy to achieve by the skilled artisan amounts to hindsight reconstruction, which is legally impermissible.

With the resubmission of the second Declaration under 37 CFR §1.132, it is clear that the pending claim is not obvious over the combination of U1, V1 and W1. Applicants respectfully request withdrawal of the rejection of claim 12.

Petition for Extension of Time/Fees Payable

Applicants hereby petition for a two (2) month extension of time, extending the deadline for responding to the June 15, 2015 Office Action from September 15, 2015 to November 15, 2015. As such, the fee of \$300.00 specified in 37 C.F.R. §1.17(a)(2) for such two (2) month extension for a

small entity is due.

The total fee of \$300.00 is being paid using Electronic Funds Transmission. Authorization is also hereby given to charge any deficiency in applicable fees, or credit any overcharges, for this response to Deposit Account No. 13-4365 of Moore & Van Allen PLLC.

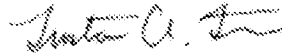
Conclusion

In view of the amended claims and the remarks presented above, it is respectfully submitted that all of the present claims of the application are in condition for immediate allowance. It is therefore respectfully requested that a Notice of Allowance be issued.

If any additional issues remain, the Examiner is requested to contact the undersigned attorney at (919) 286-8000 to discuss same.

Respectfully submitted,

MOORE & VAN ALLEN PLLC



Date: November 13, 2015

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APPENDIX A